

Project Management – Part 3

Revision Sheet

Based on: Anne Rinaldi & JJ Aureglia – *Managing Global Projects: An Approach to Management* (slides 165–244)

1 Cost Estimating

1.1 Estimate Refinements and Accuracy

Definition 1.1 (Estimate Refinement Levels). Cost and duration estimates are progressively refined through three levels as project information matures:

1. **Order of Magnitude / Top-Down (Preliminary):** Produced before or at project start using expert judgment on a high-level project description or analogous estimating.
Accuracy: -25% to $+75\%$.
2. **Budget / Conceptual / Design / Control (Planning):** Produced during planning using expert judgment on the WBS or parametric models.
Accuracy: -10% to $+25\%$.
3. **Definitive / Bottom-Up / Engineering (Refinement):** Performed by the people who will do the work, relying on detailed expert judgment.
Accuracy: -5% to $+10\%$.

Important Remark

As project detail increases, estimates become more accurate but also more costly to produce. Early rough estimates are essential for go/no-go decisions; definitive estimates are needed for baseline control.

1.2 PERT Estimating

Definition 1.2 (PERT – Program Evaluation and Review Technique). A statistical technique to improve the accuracy of activity duration (or cost) estimates by combining multiple expert opinions.

Procedure:

1. Select **several independent experts** (minimum 3).
2. Each expert delivers an independent estimate.
3. Collaboratively build a *Most Likely* estimate.
4. Identify: Optimistic (O), Most Likely (M), Pessimistic (P).

PERT Formulas

$$E_{\text{PERT}} = \frac{P + 4M + O}{6}$$
$$\sigma_{\text{PERT}} = \frac{P - O}{6}$$

Interpretation: Approximately 67% of outcomes fall within $E_{\text{PERT}} \pm \sigma_{\text{PERT}}$.

Example 1.1. Suppose three experts provide: $O = 4$ days, $M = 6$ days, $P = 14$ days.

$$E_{\text{PERT}} = \frac{14 + 4 \times 6 + 4}{6} = \frac{42}{6} = 7 \text{ days}$$

$$\sigma = \frac{14 - 4}{6} \approx 1.67 \text{ days}$$

So 67% confidence interval: [5.33, 8.67] days.

1.3 Cost Estimating Process

Definition 1.3 (Cost Estimating). The process of developing a quantitative assessment (estimate) of the costs of the resources needed to complete project activities, for each Work Package.

Cost Estimate Structure

$$\text{Cost} = \underbrace{\text{Cost of Material}}_{\text{reliable}} + \underbrace{\text{Cost of Tools} \times \text{Duration}}_{\text{challenging}} + \underbrace{\text{Cost of HR} \times \text{Duration}}_{\text{challenging}}$$

Tools for cost estimating:

- **Analogous estimating:** Use actual cost of a similar previous project as a basis.
- **Parametric modeling:** Use project characteristics (parameters) in a mathematical model.
- **Bottom-up estimating:** Estimate each work package individually, then aggregate.

Outputs: Cost Estimates, Cost Management Plan.

Common Pitfall / Warning

Cost $\neq$ Price. Cost is the amount spent on resources. Price is a business decision depending on strategy, market conditions, etc. Never confuse the two in project management.

Key Takeaways

- Three levels of estimate accuracy: Order of magnitude ($-25/+75\%$), Budget ($-10/+25\%$), Definitive ($-5/+10\%$).
- PERT formula: $E = (P + 4M + O)/6$; standard deviation: $\sigma = (P - O)/6$.
- Cost estimation difficulty lies primarily in *duration estimates*.
- Cost estimates are refined iteratively as more detail becomes available.

2 Schedule Development

2.1 Activity Sequencing

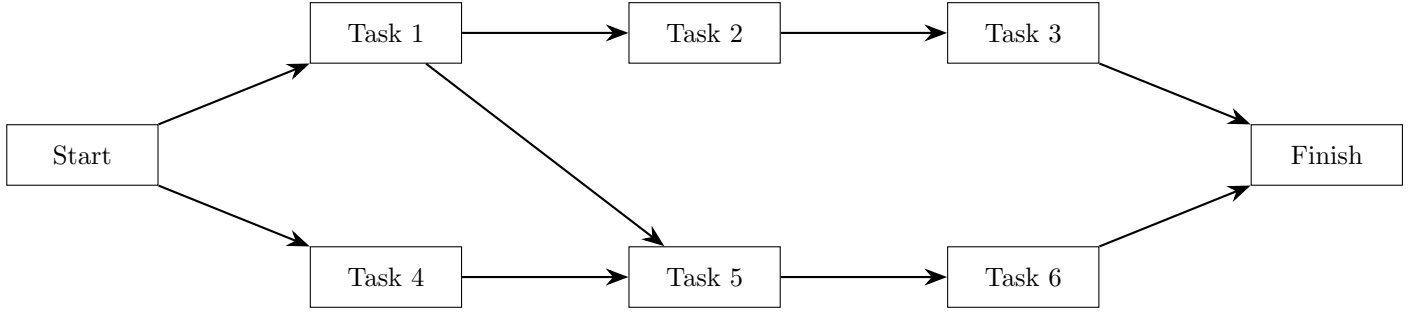
Definition 2.1 (Activity Sequencing). The process of identifying and documenting interactivity dependencies between project activities.

Definition 2.2 (Dependency Types). Four types of logical relationships (finish-to-start is the most common):

1. **Finish-to-Start (FS):** Successor starts after predecessor finishes.
2. **Finish-to-Finish (FF):** Successor finishes when predecessor finishes.
3. **Start-to-Start (SS):** Successor starts when predecessor starts.
4. **Start-to-Finish (SF):** Successor finishes when predecessor starts.

2.2 Precedence Diagramming Method (PDM)

Definition 2.3 (Precedence Diagramming Method). A network diagramming technique in which activities are represented by **nodes** (boxes) and dependencies by **arrows**. The output is the **Project Network Diagram (PND)**.



2.3 Critical Path Method (CPM)

Definition 2.4 (Critical Path Method). An algorithm that determines the **longest path** through a project network diagram, thereby determining the **minimum project duration**. Activities on this path have **zero float**.

Definition 2.5 (Key Dates for Each Activity). Each activity is characterized by four dates:

- **Early Start (ES)**: Earliest possible start date.
- **Early Finish (EF)**: Earliest possible finish date.
- **Late Start (LS)**: Latest start date without delaying the project.
- **Late Finish (LF)**: Latest finish date without delaying the project.

Activity Duration

ES	EF
LS	LF

2.3.1 Forward Pass (Computing Early Dates)

Property 2.1 (Forward Pass Rules). Starting from the **Start** node (set $ES = EF = 0$):

$$ES_i = \max(EF_j \mid j \text{ is a predecessor of } i) \quad (1)$$

$$EF_i = ES_i + d_i \quad (2)$$

where d_i is the duration of activity i .

2.3.2 Backward Pass (Computing Late Dates)

Property 2.2 (Backward Pass Rules). Starting from the **Finish** node (set $LS = ES$ of the **Finish** node):

$$LF_i = \min(LS_j \mid j \text{ is a successor of } i) \quad (3)$$

$$LS_i = LF_i - d_i \quad (4)$$

2.3.3 Float (Slack)

Definition 2.6 (Total Float). The amount of time an activity can be delayed **without delaying the project end date**:

$$\text{Float}_i = LS_i - ES_i = LF_i - EF_i$$

- $\text{Float} > 0$: schedule flexibility exists.
- $\text{Float} = 0$: the activity is **critical**.

Definition 2.7 (Free Float). The amount of time an activity can be delayed **without delaying the early start of any successor activity**.

Definition 2.8 (Critical Path). The sequence of activities with **float = 0**. It determines the overall project duration. Any delay on a critical-path activity delays the entire project.

Example 2.1 (Project Epsilon – CPM Walkthrough). Given the network with durations: Task 1 = 2, Task 2 = 4, Task 3 = 3, Task 4 = 7, Task 5 = 1, Task 6 = 5.

Dependencies: Start → T1, T4; T1 → T2, T5; T4 → T5; T2 → T3; T5 → T6; T3, T6 → Finish.

Task	Duration	ES	EF	LS	LF	Float
Task 1	2	0	2	1	3	1
Task 2	4	2	6	3	7	1
Task 3	3	6	9	7	10	1
Task 4	7	0	7	0	7	0
Task 5	1	7	8	9	10	2
Task 6	5	7	10	5	10	—

Note: Looking at the slides' values more carefully:

- T1: ES=0, EF=2, LS=1, LF=3, Float=1
- T2: ES=2, EF=6, LS=3, LF=7, Float=1
- T3: ES=6, EF=9, LS=7, LF=10, Float=1 (not on critical path in this configuration)
- T4: ES=0, EF=7, LS=0, LF=7, Float=0 ← Critical
- T5: ES=7, EF=8, LS=9, LF=10, Float=2
- T6: ES=5, EF=10, LS=5, LF=10, Float=0 ← Critical

Critical Path: Start → Task 4 → Task 6 → Finish. **Project Duration = 10 days.** (Note: the exact topology depends on additional links shown in slides; Task 6 connects after Task 2 finishes based on ES=5.)

From slide values: Float(T1)=1, Float(T2)=2, Float(T3)=5, Float(T4)=0, Float(T5)=1, Float(T6)=0.

Critical Path: Start → T4 → T6 → Finish (duration = 7 + 5 = 12? No, project duration = 10 per slide). The critical path is derived by activities with float = 0.

Important Remark

The GANTT chart is the standard **schedule representation** derived from the CPM. It shows activities as horizontal bars plotted against time, making the schedule visually intuitive.

2.4 Duration Compression

Definition 2.9 (Duration Compression). Techniques to shorten the project schedule **without changing the project scope**.

Definition 2.10 (Crashing). Decreasing the duration of activities **on the critical path** by applying **additional resources**.

- Incurs **additional costs** due to lower efficiency.
- Strategy: crash activities that yield the **most time savings per dollar spent**.
- Source of **cost overrun**.

Definition 2.11 (Fast Tracking). Performing normally **sequential** activities **in parallel** (or partially overlapping).

- Requires **redefining logical relationships** between activities.
- Often requires refining the WBS and Activity Lists.
- Considered **less costly** than crashing.
- Introduces **additional risks** due to potential rework (work starts before precise information from predecessors is available).

Example 2.2. Starting coding before design is complete (fast tracking). Sending early code drops to the testing team before all code is finished.

Common Pitfall / Warning

Fast tracking increases risk of rework. Only apply it when schedule compression is critical and the risk is acceptable. Always document the rationale and associated risks.

Key Takeaways

- CPM uses forward pass (ES, EF) and backward pass (LS, LF) to identify the critical path.
- **Float** = **LS** – **ES** = **LF** – **EF**; activities with float = 0 are critical.
- **Crashing** adds resources to critical-path activities $\Rightarrow$ higher cost.
- **Fast tracking** overlaps sequential tasks $\Rightarrow$ higher risk but lower cost than crashing.
- In Agile, the project-level schedule is less relevant; focus is on sprint schedules.

3 Cost Budgeting and Baseline

3.1 Cost Budgeting

Definition 3.1 (Cost Budgeting). The process of merging **scope, cost, and schedule** information to establish a **cost baseline** against which project performance will be measured.

Definition 3.2 (Cost Baseline (S-Curve)). A cumulative S-curve plotting **cumulative planned costs as a function of time**. Each point represents how much value is planned to be delivered at that moment.

Cost accrual methods:

- **Recommended:** Accrue 100% of the cost when the task is planned to be *completed*.
- **Alternative (50/50 rule):** Accrue 50% when the task starts, remaining 50% upon completion.

Definition 3.3 (BAC – Budget At Completion). The **total planned budget** for the project at time $t = 0$. It is the final value on the cost baseline curve.

$$\text{BAC} = \sum_{\text{all WP}} \text{Budgeted Cost of Work Package}$$

Example 3.1 (Project Epsilon Cost Baseline). With 1 person per task at \$1000/day:

- End of Task 4 (day 7): cumulative = \$7,000
- End of Task 1 (day 2): cumulative = \$2,000
- Progressive accumulation forms the S-curve
- BAC = \$22,000

The % complete at any time t is: $\frac{\text{Cost Baseline}(t)}{\text{BAC}} \times 100\%$.

Important Remark

Cost is treated as a proxy for **value delivered**: as each work package is completed, value is delivered to the project. This is a fundamental assumption in Earned Value Analysis.

3.2 Project Management Plan

Definition 3.4 (Project Management Plan). The consolidation of **all planning outputs**: scope (WBS), cost estimates, schedules, cost baseline, communication plan, quality plan, risk management plan, and key decisions. It serves three purposes:

1. Guide project execution.
2. Facilitate communication among stakeholders.
3. Provide a baseline for progress measurement and project control.

Definition 3.5 (PMIS – Project Management Information System). Tools and techniques (manual and automated) used to gather, integrate, and disseminate the outputs of project management processes.

Key Takeaways

- Cost budgeting merges scope + cost + schedule into the **cost baseline** (S-curve).
- $BAC = \text{total planned project cost} = \text{final point on the S-curve}$.
- The cost baseline is the **reference** for all performance measurement (EVA).
- The Project Management Plan consolidates all planning outputs into one reference document.

4 Execution Management and Control

4.1 Project Launching

Definition 4.1 (Project Execution). The phase during which the product of the project is actually created. Work starts and charges to the project begin occurring.

Key elements of launching:

- **Work Authorization System:** Formal procedure for sanctioning project work to ensure it is done at the right time and in proper sequence.
- **Project Kick-Off Meeting:** Builds team buy-in and morale.
- **Phase Gates:** Opportunities to celebrate completion or stop the project.
- **Resource Availability:** Anticipate and verify timely resource availability *ahead of time*.

4.2 Performance Reporting

Definition 4.2 (Performance Reporting). The process of gathering, exchanging, and communicating information about project progress. Key activity: verifying what has been **actually completed** (at the work package level) with expected quality.

Two channels of information:

1. **Formal:** Structured periodic meetings (weekly, daily), reviews, status reports.
2. **Informal:** Management by Wandering Around (MBWA), straight talk, alternative communication channels.

Important Remark

Track items that are **not formally measured**: team morale, confidence, stress levels. These are often leading indicators of future problems.

4.3 Earned Value Analysis (EVA)

Definition 4.3 (Earned Value Analysis). A technique that **integrates scope, cost, and schedule** to assess project performance by comparing three key values at any point in time:

1. **BCWS** (Budgeted Cost of Work Scheduled) = **Planned Value (PV)**: budget to date — “what we should have delivered.”
2. **BCWP** (Budgeted Cost of Work Performed) = **Earned Value (EV)**: budget of work actually completed — “what we have actually delivered.”
3. **ACWP** (Actual Cost of Work Performed) = **Actual Cost (AC)**: total costs actually incurred — “what we have actually spent.”

Definition 4.4 (Estimating BCWP). Two methods:

- **0/100 Rule (Recommended):** Accrue 100% of budgeted cost only when a task is *complete*; 0% otherwise.
- **50/50 Rule:** Accrue 50% when a task starts, remaining 50% upon completion. Provides a statistical approximation when many tasks are considered but **bears higher risks**. Not recommended.

4.3.1 Variance Indicators

EVA Variance Formulas

Schedule Variance:

$$SV = BCWP - BCWS$$

- $SV < 0$: schedule **delay**
- $SV > 0$: **ahead** of schedule

Cost Variance:

$$CV = BCWP - ACWP$$

- $CV < 0$: cost **overrun**
- $CV > 0$: cost **under run** (saving money)

4.3.2 Performance Indices

EVA Performance Indices

Schedule Performance Index:

$$SPI = \frac{BCWP}{BCWS}$$

- $SPI < 1$: schedule delay
- $SPI > 1$: ahead of schedule

Cost Performance Index:

$$CPI = \frac{BCWP}{ACWP}$$

- $CPI < 1$: cost overrun
- $CPI > 1$: cost under run

4.3.3 Estimate At Completion (EAC)

Definition 4.5 (EAC – Estimate At Completion). A forecast of total project costs based on current performance. Three scenarios:

EAC Formulas

Scenario 1 – Variance is typical (expected to continue):

$$EAC = \frac{BAC}{CPI}$$

CPI is shown to be relatively stable after 15–20% project completion.

Scenario 2 – Variance was exceptional (will not recur):

$$EAC = ACWP_{\text{to date}} + \text{Remaining Budget}$$

Scenario 3 – Assumptions changed (new estimate required):

$$EAC = ACWP_{\text{to date}} + \text{New Estimate for Remaining Work}$$

Example 4.1 (Project Epsilon EVA). On March 18 (end of day), with Tasks 1, 2, 4, and 6 completed. $ACWP = \$19,000$.

From the cost baseline (S-curve), the BCWS at that date corresponds to the planned cumulative value.

BCWP = budgeted costs of completed tasks:

$$BCWP = 2000 + 4000 + 7000 + 5000 = \$18,000$$

- $CV = 18,000 - 19,000 = -\$1,000 \Rightarrow$ cost overrun

- $CPI = 18,000/19,000 \approx 0.947 \Rightarrow$ for every dollar spent, only \$0.95 of value delivered
- $EAC = 22,000/0.947 \approx \$23,232$ (if variance is typical)

Important Remark

In Agile, without a traditional project schedule and cost budgeting, Earned Value Analysis is **not directly applicable**. Agile uses velocity and burndown charts instead.

4.4 Cost and Schedule Control

Definition 4.6 (Cost/Schedule Control). The processes of:

- Analyzing causes of cost/schedule variances.
- Deciding if corrective action is needed.
- Updating cost estimates (EAC) and project schedule.
- Managing changes, actions, and updates.

When justified, the Project Management Plan may be **re-baselined** with new estimates.

4.5 Anticipation Best Practices

Key anticipation strategies:

- Thorough **risk management** and detecting **symptoms** early.
- Secure commitments with periodical checking.
- Carefully manage **prerequisites**.
- Avoid **black boxes** (no visibility for long periods).
- Immediately focus on **small delays** — do not wait.
- Stay at least **one move ahead** on the critical path.

4.6 Change Management

Definition 4.7 (Change Management). A formal process to handle changes to project scope, schedule, or cost:

1. **Formal Request:** Document the proposed change.
2. **Impact Analysis:** Assess effects on cost, schedule, risks, quality.
3. **Decision:** Approve/reject via a Change Review Board.
4. **Update:** Revise the Project Management Plan accordingly.

Important Remark

In Agile, change management is embedded in the process (backlog refinement). However, ensure that **fixed scope is not contracted** — i.e., the contracted scope should not be unilaterally changed.

Key Takeaways

- EVA compares three values: BCWS (planned), BCWP (earned), ACWP (actual).
- $SV = BCWP - BCWS$; $CV = BCWP - ACWP$.
- $SPI = BCWP/BCWS$; $CPI = BCWP/ACWP$.
- $EAC = BAC/CPI$ if variance is typical.
- Change management: formal request $\rightarrow$ impact analysis $\rightarrow$ decision $\rightarrow$ update.
- Anticipate: detect symptoms, avoid black boxes, address small delays immediately.

5 Risk Management

5.1 Enterprise vs. Project Risk Management

Definition 5.1 (Enterprise Risk Management (ERM)). A comprehensive framework (ISO 31000) covering:

- **Operational Risks** (business management)
- **Financial Risks** (investment management)
- **Strategic Risks** (enterprise change management)
- **Hazard Risks** (casualty/actuarial)

Project Risk Management is a **subset** of ERM, focused on risks within a specific project.

5.2 The Notion of Risk

Definition 5.2 (Risk (Project Management)). An **uncertain event** whose outcome could affect the project. Unlike the general language definition (always negative), PM risks are of two kinds:

- **Opportunities:** Positive outcomes.
- **Threats:** Negative outcomes.

Definition 5.3 (Purpose of Risk Management). Anticipate uncertain events to ensure the project meets its targets:

- Prevent surprises (especially bad ones) — avoid management by crisis.
- Prevent problems from occurring or minimize their impact.
- Recognize opportunities before competitors.
- Maximize results of positive events, minimize consequences of adverse events.

5.3 Risk Management Process

The risk management process consists of five steps:

1. **Risk Management Planning**
2. **Risk Identification**
3. **Risk Analysis** (Qualitative and Quantitative)
4. **Risk Response Planning**
5. **Risk Monitoring and Control**

5.3.1 Risk Identification

Definition 5.4 (Risk Identification). Determining which risks are likely to affect the project and documenting their characteristics. Identify the main **sources of risk** and possible **risk events**.

Two major initial sources of risks in projects:

1. **SCOPE (WBS):** The WBS identifies all deliverables; failure to produce any deliverable is a risk.
 2. **RESOURCES:** Unavailability of resources according to the project schedule.
- Technique:** Brainstorming (creativity is needed).

5.3.2 Risk Analysis

Definition 5.5 (Risk Analysis Parameters). For each identified risk event, estimate:

- **Probability of occurrence (P):** may use historical data, company statistics.
- **Impact / Range of outcomes (I):** cost of the work package at risk, contract penalties, lost revenue, or qualitative rating.
- **Expected timing and frequency.**

- **Detection indicators (risk symptoms):** indirect manifestations (e.g., cost overruns may indicate poor estimating; low morale may indicate difficulties).

Expected Monetary Value (EMV)

$$EMV = P \times I$$

where P = probability of the risk event and I = monetary impact (value).

EMVs can be aggregated to calculate a **range of total project costs** or define **contingencies**.

Definition 5.6 (Risk Classification for Response). From the list of identified risk events, classify each as:

- **Opportunities to pursue** (warrant response)
- **Opportunities to ignore**
- **Threats to respond to** (warrant response)
- **Threats to accept**

For ignored opportunities and accepted threats: **document the decision and who made it**.

Example 5.1 (Risk Priority Matrix).

Risk Event	Probability	Impact	Respond?
Risk Event 1	High	High	Yes
Risk Event 2	High	Medium	Yes
Risk Event 3	Medium	High	Yes
Risk Event 4	Medium	Medium	Yes
Risk Event 5	Medium	Low	No
Risk Event 6	Low	Medium	No
Risk Event 7	Low	Low	No

5.3.3 Risk Response Planning

Definition 5.7 (Risk Response Strategies for Threats). Three categories:

1. **Avoidance:** Eliminate the threat by changing the project plan (e.g., change the WBS, adopt a safer design approach).
2. **Mitigation:** Reduce the **probability** or **impact** of the risk event (e.g., plan contingencies, add safety margins).
3. **Acceptance:** Acknowledge the risk and take no proactive action; deal with it if it occurs.

Example 5.2 (Electronic Card Design). **Risk:** One-pass design for electronic cards bears high risk of rework if a bug is found late.

- **Avoidance:** Switch to a two-pass design (longer but safer). Impacts: larger scope, longer schedule, higher cost, better quality.
- **Mitigation:** Keep one-pass design but plan physical layout with external connection points for quick fixes. More costly but saves time vs. two-pass.
- **Acceptance:** Keep one-pass design to meet customer deadlines; assign best engineers.

Example 5.3 (Car Travel Risk). Source of risk: travelling by car to work.

Risk event: accident.

- **Avoidance:** Walk to work, take public transport, or work from home.
- **Mitigation:** Insurance, advanced driving lessons, buy a safe car, always fasten seatbelt.
- **Acceptance:** No specific action.

Definition 5.8 (Risk Response Planning Output). Two types of planned actions:

1. **Immediate actions:** Integrated into the WBS and Project Management Plan (preventive measures).
2. **Contingency actions:** Executed only if and when the risk event occurs.

The **Risk Management Plan** documents: procedures, risk identification/analysis results, responsibilities, and responses developed.

5.3.4 Risk Monitoring and Control

Definition 5.9 (Risk Monitoring and Control). The process of:

- Detecting risk occurrence.
- Executing planned risk responses.
- Analyzing changes in risk over the project lifecycle.
- Repeating the identify → analyze → respond cycle when changes occur.

Definition 5.10 (Workaround). An **unplanned response** to a negative risk event that was not anticipated in the risk management plan.

When an unplanned event occurs:

- Avoid panic; avoid blame.
- Look for creativity: brainstorming, out-of-the-box thinking.
- Get help from management or the client as needed.

Key Takeaways

- Risks = **opportunities** (positive) + **threats** (negative).
- Risk process: Plan → Identify → Analyze → Respond → Monitor & Control.
- Two major risk sources in projects: **Scope (WBS)** and **Resources**.
- $EMV = P \times I$ for quantitative risk assessment.
- Three response strategies for threats: **Avoidance, Mitigation, Acceptance**.
- A **workaround** is an unplanned response to an unanticipated risk.
- Always document accepted risks and who made the decision.

6 Quality Management

6.1 Why Quality?

Quality is a **business matter**:

1. **Win & Keep Clients**: Meet client expectations; quality certifications as a marketing weapon.
2. **Optimize Costs to Optimize Profit**: Total Cost of Quality; defect *prevention* vs. defect *correction*.

Definition 6.1 (Quality (ISO 9000)). The ability of a product, system, or process (set of characteristics) to satisfy **customer and other stakeholders' requirements**.

$$\text{Quality} = \text{Satisfying Client Needs at the Best Cost}$$

6.2 Two Approaches to Quality

Definition 6.2 (Quality Assurance (QA)). **Short-term** approach pursuing conformance to specifications and satisfying customer needs. A set of **predefined actions** providing confidence that a product/service will satisfy requirements.

Definition 6.3 (Continuous Improvement). **Long-term** approach based on improving the application area process. Pursues **excellence** and **value attributes**.

6.3 Planning for Quality

The Quality Plan includes:

1. Methods used.
2. QA reviews performed (result in actions → WBS items → scope).
3. Continuous improvement by injecting lessons learned.
4. Mindset through education.

6.4 The Four Quality Types

Definition 6.4 (Quality vs. Grade). Quality is often opposed to Grade. **Low quality is always a problem; low grade may not be.** Zero defect $\neq$ value.

6.4.1 1. Conformance to Specifications (Provider Logic)

- Development of standards (public and customer specifications).
- Conformance must be **verified** through quality control.
- Examples: interoperability tests, homologation tests.
- Reliability is an important criterion for **sustainability**.

6.4.2 2. Excellence (Provider Logic)

- Quest for perfection (zero defect, Six Sigma).
- Originated from craftsmanship; quality problems appeared with industry/volume production.
- Think of **economical constraints**: no gold plating.
- Reduce the **Total Cost of Quality**: prevention vs. correction.

Common Pitfall / Warning

No Gold Plating: Do not add features or quality beyond what was specified. Use Value Analysis to determine what is worth investing in.

6.4.3 3. Satisfying Customer Expectations (Customer Logic)

- Most common quality definition today (driven by the growth of the services market).
- Some expectations are hard to capture in written specifications.
- Strategies: maximize formal capture, think of **implied** (not explicit) expectations, periodically verify with client (prototypes, etc.).

6.4.4 4. Value Attribute (Customer Logic)

Definition 6.5 (Value Attribute). Quality viewed through the **price/quality ratio**: what the customer gets for what they pay.

Definition 6.6 (Value Analysis / Value Engineering). A product is seen as a **set of functions** (vs. a set of components). Process:

1. **Functional Analysis:** Understand client needs (explicit and implicit), understand competition.
2. For each function, estimate **client value** and **cost**.
3. **Prioritize** to establish functional specifications.

Definition 6.7 (Value Management / Customer Value Management). A complete business management system that:

- Understands customer purchase and repurchase behaviors.
- Adapts products and processes accordingly.

6.5 Quality Logics Summary

	Customer Logic	Provider Logic
QA (Short Term)	Satisfying Customer Needs	Conformance to Specifications
CI (Long Term)	Value Attributes	Excellence

6.6 Main Tools for Quality

Definition 6.8 (Quality Assurance Reviews). **Independent** review activities performed by specially trained, authorized individuals. QA reviews include:

- Ensuring high-quality, profitable solutions.
- Recognition, containment, and mitigation of risks.
- Sustainability reviews may also be included.

Definition 6.9 (Continuous Process Improvement).
1. Document formal processes.
2. Continuously examine performance.
3. Update business processes to: proliferate best practices, correct defects.

6.7 Quality Hints (Best Practices)

- Think of quality **from the customer's point of view**.
- Turn implied needs into stated needs through **scope management**.
- Prioritize functions to deliver **value**.
- Quality $\neq$ Grade: low quality is always a problem.
- Zero defect does NOT mean you have value.
- Project Quality Management addresses both the **quality of the project** and the **product of the project**.
- Use **independent reviews**.
- Continuously improve by learning (**Lessons Learned**).

Important Remark

Quality Assurance often performs Risk Management. The purpose of QA reviews includes recognition, containment, and mitigation of risks that could jeopardize project success.

Key Takeaways

- Four quality types: Conformance to Specs, Excellence, Customer Expectations, Value Attribute.
- Quality $\neq$ Grade; low quality is always a problem.
- Two logics: Customer (expectations, value) vs. Provider (specs, excellence).
- Two approaches: QA (short-term) vs. Continuous Improvement (long-term).
- Value Analysis: view product as functions, prioritize by client value/cost ratio.
- QA reviews must be **independent** and may include risk and sustainability assessments.

7 Project Closing and Conclusion

7.1 Closing the Project

Definition 7.1 (Project Closure). Every project requires formal closure, which involves:

- **Administrative closure + Contract close-out** (procurement).
- Verifying and documenting project results.
- **Formal acceptance**: Documentation that the client/sponsor has accepted the product.
- Documenting **Lessons Learned** for continuous improvement.

Definition 7.2 (Administrative Closure Checklist).
• All invoicing has been issued.

- No pending change requests (any pending must be reviewed).
- All project-related costs have been claimed/transferred to contract:
 - Labor costs
 - Travel costs
 - Asset-related costs
 - Subcontractor-related costs

7.2 Communication and Management Tips

Best practices for project managers:

- Set priorities and address high-priority items first.
- Manage stakeholders' expectations actively.
- Meeting minutes: set baseline, document decisions, assign actions.
- Write key agreements formally.
- Plan for communication explicitly.
- Manage stress and workload.
- **Fast start:** the schedule battle is often lost in the early days.
- Thorough planning and thorough change management.
- Verify status through **both formal and informal** channels.

7.3 Root Causes of Troubled Projects

Definition 7.3 (Root Causes – Design/Proposal Phase). • Inaccurate project estimates.

- Failure to set and manage customer expectations.
- Overlooking stakeholders' expectations management.
- Failure to establish an appropriate contractual baseline.
- Failure to reach a common understanding of requirements or completion criteria.
- Failure to plan for risk containment.
- Concessions during negotiations with no price adjustment.

Definition 7.4 (Root Causes – Delivery Phase). • Lack of or inadequate project management discipline.

- Ineffective financial management.
- Failure to implement a proper change management process.
- Lack of or inadequate project management plan/schedule.
- Inability to acquire properly skilled resources.
- Unfulfilled customer responsibilities.
- Overlooking stakeholders' expectations management.

Key Takeaways

- Every project needs formal **administrative closure** and **formal acceptance**.
- Document **Lessons Learned** systematically for organizational learning.
- Most troubled projects fail due to: poor estimates, unmanaged expectations, inadequate change management, and insufficient planning discipline.
- The schedule battle is often lost in the **early days** — fast start is critical.
- Verify status through **both formal and informal channels**.